Mikhail Razakov

Quant Developer / Research Engineer | C++ · Python

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I build algorithmic program analyses and reproducible research tooling in C++ and Python. PS-form combines exact affine counting with modular proofs; independent reference oracles and metamorphic tests check the results.

Engineering context

ISP RAS: SharpChecker symbolic computation · MCST: LLVM 22/C++23.

Algorithms & research

PS-form: conservative solver, exact oracle, randomized/metamorphic validation; UniversalToolchain: controlled verifier experiment.

C++ / Python research tooling

LLVM, Python harnesses, graph algorithms, differential execution, reproducible counterexamples, and test oracles.

EXPERIENCE

October 2026 — present

ISP RAS — SharpChecker development

- Work on symbolic computation in SharpChecker, a C#/.NET static-analysis tool.

July 1 — August 31, 2026

MCST — Compiler Engineering Intern

- Implemented an LLVM LICM pass in C++; transformation legality is derived from loop structure, memory accesses, side effects, and speculative-safety constraints.
- Built a Python harness comparing execution before/after optimization and checking explicit transformation / no-transformation cases; also implemented RPO, Dijkstra, Dinic, and Tarjan SCC in C++23.

PROJECTS

PS-form Memory Dependence Analyzer

Exact affine iteration-pair counting and modular/GCD disjointness proofs; yes/no cross-checked with a small-domain oracle and metamorphic tests, unresolved cases remain maybe.

LLVM Interprocedural Global IV Optimization

A conservative LLVM 22 prototype/MVP with an actual IR transform; 29 positive/negative regression cases cover verifier validity, idempotence, and before/after execution.

UniversalToolchain / Wist2

Implemented an end-to-end pipeline with frontend, IR, optimization passes, interpreter, and a CIL backend; current exact test manifest: 1,324/1,324.

SKILLS

Programming

C++23, Python, C17, C#, Rust

Algorithms & analysis

data structures, graph algorithms, CFG/SSA, program analysis, memory dependence

Research engineering

exact/reference oracles, differential/randomized/metamorphic testing, reproducible harnesses, counterexamples

Systems

LLVM 22, x86-64, Linux, CMake, generated-code analysis

EDUCATION

HSE University — Software Engineering

RECOGNITION

Organizer correspondence: only 5.0/5.0 score and only solution passing all 104/104 tests among 49 submissions.

Moscow / remote